SSU-Centric Topology Analysis Report

Generated at: 2026-04-16 16:21:19

This report follows the same dark, topology-first presentation as the HTML dashboard. It compares the selected topologies under the shared 8 SSU + 2 Union exchange-node model: 2D-FullMesh, 2D-FullMesh 2x4, 2D-Torus, 2D-Torus Best Twist, 3D-Torus, 3D-Torus Best Twist, 3D-Torus 2x4x3, 3D-Torus 2x4x3 Best Twist, 3D-Torus 2x4x2, 3D-Torus 2x4x2 Best Twist, 3D-Torus 2x4x1, 3D-Torus 2x4x1 Best Twist, Clos, Dragon-Fly, SparseMesh S=5;N=2, SparseMesh S=3;N=4.

Topology Snapshot

Topology	Diameter	Average Hops	Bisection BW (Gbps)	Bisection BW / SSU	Default Route Throughput (Gbps)	Sparse P95 Completion (ms)
2D-FullMesh	4	3.512	12,800.00	200	379.1	0.24
2D-FullMesh 2x4	4	3.27	3,200.00	100	196.9	0.18
2D-Torus	5	3.524	3,200.00	100	131.2	0.18
2D-Torus Best Twist	4	3.27	6,400.00	200	315	0.18
3D-Torus	7	4.51	12,800.00	100	132.8	0.16
3D-Torus Best Twist	5	4.071	25,600.00	200	289.8	0.16
3D-Torus 2x4x3	6	4.178	9,600.00	100	132.6	0.16
3D-Torus 2x4x3 Best Twist	5	3.927	19,200.00	200	298.4	0.16
3D-Torus 2x4x2	6	4.016	6,400.00	100	132.3	0.24
3D-Torus 2x4x2 Best Twist	4	3.512	12,800.00	200	328.4	0.24
3D-Torus 2x4x1	5	3.524	3,200.00	100	131.2	0.18
3D-Torus 2x4x1 Best Twist	4	3.27	6,400.00	200	315	0.18
Clos	4	3.902	28,800.00	400	400	0.16
Dragon-Fly	5	4.333	33,600.00	161.5	199.5	0.16
SparseMesh S=5;N=2	4	3.287	9,600.00	218.2	362.5	0.24
SparseMesh S=3;N=4	4	3.398	12,800.00	246.2	371.4	0.24

2D-FullMesh

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8

Field	Value
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	4x4 exchange nodes
Exchange Node Count	16
Ssu Count	128
Union Count	32
Internal Link Count	256
Backend Link Count	96
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one 4x4 full-mesh plane: 3 row links on X and 3 column li

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Routing Diversity Snapshot

SHORTEST_PATH vs DOR: 1.60x average path diversity (3.20 vs 2.00 end-to-end paths per inter-exchange SSU pair); FULL_PATH reaches 12.00 on average with peak 12 paths.

Mode	Avg Paths	Peak Paths
DOR	2	2
SHORTEST_PATH	3.2	4
FULL_PATH	12	12

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	DOR Throughput	379.10
Sparse M-to-N	DOR Throughput	200.00

Structural Metrics

Metric	Value
Diameter	4
Average Hops	3.512
Bisection Bandwidth Gbps	12,800.00
Bisection Bandwidth Gbps Per Ssu	200

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	379.10 Gbps	0.0107 s	0.0107 s	3.51	95.52%	0.000
SHORTEST_PATH	379.10 Gbps	0.0107 s	0.0107 s	3.51	95.52%	0.000
FULL_PATH	190.50 Gbps	0.0213 s	0.0107 s	4.39	100.00%	0.167

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	200.00 Gbps	0.0003 s	0.0002 s	3.53	62.50%	1.216
SHORTEST_PATH	200.00 Gbps	0.0003 s	0.0002 s	3.53	43.75%	0.874
FULL_PATH	200.00 Gbps	0.0003 s	0.0002 s	4.35	43.75%	0.458

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- both Union chips expose the same direct-connect 4x4 full-mesh plane, so routing decisions are evaluated independently on each Union plane
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 379.10 Gbps with maximum backend link utilization 95.52%.
- Sparse M-to-N communication completes in 0.24 ms at p95 with link-utilization CV 0.874.
- Structural span is diameter 4 and average hops 3.51.

2D-FullMesh 2x4

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2x4 exchange nodes
Exchange Node Count	8
Ssu Count	64
Union Count	16
Internal Link Count	128
Backend Link Count	32
Backend Ports Per Union	4
Backend Pattern	Each Union uses 4 backend ports on one 2x4 full-mesh plane: 3 row links on X and 1 column li

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Routing Diversity Snapshot

SHORTEST_PATH vs DOR: 1.43x average path diversity (2.86 vs 2.00 end-to-end paths per inter-exchange SSU pair); FULL_PATH reaches 8.00 on average with peak 8 paths.

Mode	Avg Paths	Peak Paths
DOR	2	2

Mode	Avg Paths	Peak Paths
SHORTEST_PATH	2.857	4
FULL_PATH	8	8

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	DOR Throughput	196.87
Sparse M-to-N	DOR Throughput	266.67

Structural Metrics

Metric	Value
Diameter	4
Average Hops	3.27
Bisection Bandwidth Gbps	3,200.00
Bisection Bandwidth Gbps Per Ssu	100

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	196.87 Gbps	0.0102 s	0.0056 s	3.27	100.00%	0.346
SHORTEST_PATH	196.87 Gbps	0.0102 s	0.0056 s	3.27	100.00%	0.346
FULL_PATH	112.50 Gbps	0.0179 s	0.0056 s	4.03	100.00%	0.248

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	266.67 Gbps	0.0002 s	0.0002 s	3.28	50.00%	0.810
SHORTEST_PATH	266.67 Gbps	0.0002 s	0.0002 s	3.28	58.33%	0.780
FULL_PATH	266.67 Gbps	0.0002 s	0.0002 s	4.14	70.83%	0.338

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- both Union chips expose the same direct-connect 2x4 full-mesh plane, so routing decisions are evaluated independently on each Union plane
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order

- A2A per-SSU throughput reaches 196.87 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.18 ms at p95 with link-utilization CV 0.780.
- Structural span is diameter 4 and average hops 3.27.

2D-Torus

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4 torus planes | 16 Union | 64 SSU
Exchange Node Count	8
Ssu Count	64
Union Count	16
Internal Link Count	128
Backend Link Count	32
Backend Ports Per Union	4
Backend Pattern	Each Union uses 4 backend ports on one 2x4 torus plane, and the two Union chips inside each

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Routing Diversity Snapshot

SHORTEST_PATH vs DOR: 2.14x average path diversity (4.29 vs 2.00 end-to-end paths per inter-exchange SSU pair); FULL_PATH reaches 6.00 on average with peak 6 paths.

Mode	Avg Paths	Peak Paths
DOR	2	2
SHORTEST_PATH	4.286	12
FULL_PATH	6	6

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	DOR Throughput	131.25
Sparse M-to-N	DOR Throughput	228.57

Structural Metrics

Metric	Value
Diameter	5
Average Hops	3.524
Bisection Bandwidth Gbps	3,200.00
Bisection Bandwidth Gbps Per Ssu	100

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	131.25 Gbps	0.0154 s	0.0056 s	3.52	100.00%	0.577
SHORTEST_PATH	196.87 Gbps	0.0102 s	0.0056 s	3.52	100.00%	0.333
FULL_PATH	102.72 Gbps	0.0196 s	0.0056 s	4.29	100.00%	0.376

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	228.57 Gbps	0.0003 s	0.0002 s	3.72	100.00%	1.079
SHORTEST_PATH	266.67 Gbps	0.0002 s	0.0002 s	3.72	91.67%	0.721
FULL_PATH	252.63 Gbps	0.0003 s	0.0002 s	4.51	100.00%	0.521

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU

- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4 torus planes, so routing evaluates the two planes separately
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 196.87 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.18 ms at p95 with link-utilization CV 0.721.
- Structural span is diameter 5 and average hops 3.52.

2D-Torus Best Twist

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4 torus planes | 16 Union | 64 SSU | axis0=[0, 2] | axis1=[0, 0]
Exchange Node Count	8
Ssu Count	64
Union Count	16
Internal Link Count	128
Backend Link Count	32
Backend Ports Per Union	4
Backend Pattern	Each Union uses 4 backend ports on one 2x4 torus plane, with a fixed best-twist wrap offset tha

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH

Field	Value
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	SHORTEST_PATH Throughput	315.00
Sparse M-to-N	SHORTEST_PATH Throughput	266.67

Structural Metrics

Metric	Value
Diameter	4
Average Hops	3.27
Bisection Bandwidth Gbps	6,400.00
Bisection Bandwidth Gbps Per Ssu	200

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	315.00 Gbps	0.0064 s	0.0056 s	3.27	100.00%	0.000
FULL_PATH	112.50 Gbps	0.0179 s	0.0056 s	4.03	100.00%	0.312

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	266.67 Gbps	0.0002 s	0.0002 s	3.41	58.33%	0.586
FULL_PATH	266.67 Gbps	0.0002 s	0.0002 s	4.20	79.17%	0.453

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4 torus planes, so routing evaluates the two planes separately
- the wrap edges use a fixed best-twist offset, so deterministic DOR is intentionally not exposed for this topology

- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 315.00 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.18 ms at p95 with link-utilization CV 0.586.
- Structural span is diameter 4 and average hops 3.27.

3D-Torus

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4x4 torus planes | 64 Union | 256 SSU
Exchange Node Count	32
Ssu Count	256
Union Count	64
Internal Link Count	512
Backend Link Count	192
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one 2x4x4 torus plane, and the two Union chips inside ea

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25

Field	Value
Sparse Target Count	2

Routing Diversity Snapshot

SHORTEST_PATH vs DOR: 10.48x average path diversity (20.97 vs 2.00 end-to-end paths per inter-exchange SSU pair); FULL_PATH reaches 10.00 on average with peak 10 paths.

Mode	Avg Paths	Peak Paths
DOR	2	2
SHORTEST_PATH	20.97	240
FULL_PATH	10	10

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	DOR Throughput	132.81
Sparse M-to-N	DOR Throughput	200.00

Structural Metrics

Metric	Value
Diameter	7
Average Hops	4.51
Bisection Bandwidth Gbps	12,800.00
Bisection Bandwidth Gbps Per Ssu	100

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	132.81 Gbps	0.0614 s	0.0210 s	4.51	100.00%	0.566
SHORTEST_PATH	199.22 Gbps	0.0410 s	0.0210 s	4.51	100.00%	0.283
FULL_PATH	58.81 Gbps	0.1388 s	0.0210 s	5.45	100.00%	0.516

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	200.00 Gbps	0.0003 s	0.0002 s	4.53	87.50%	0.918
SHORTEST_PATH	200.00 Gbps	0.0003 s	0.0002 s	4.53	54.58%	0.468
FULL_PATH	200.00 Gbps	0.0003 s	0.0002 s	5.50	95.00%	0.614

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching

- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4x4 torus planes, so routing evaluates the two planes separately
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 199.22 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.16 ms at p95 with link-utilization CV 0.468.
- Structural span is diameter 7 and average hops 4.51.

3D-Torus Best Twist

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4x4 torus planes | 64 Union | 256 SSU | axis0=[0, 2, 2] | axis1=[0, 0, 0] | axis2
Exchange Node Count	32
Ssu Count	256
Union Count	64
Internal Link Count	512
Backend Link Count	192
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one 2x4x4 torus plane, with a fixed best-twist wrap offset

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	SHORTEST_PATH Throughput	289.77
Sparse M-to-N	SHORTEST_PATH Throughput	200.00

Structural Metrics

Metric	Value
Diameter	5
Average Hops	4.071
Bisection Bandwidth Gbps	25,600.00
Bisection Bandwidth Gbps Per Ssu	200

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	289.77 Gbps	0.0282 s	0.0210 s	4.07	100.00%	0.000
FULL_PATH	75.59 Gbps	0.1079 s	0.0210 s	5.01	100.00%	0.384

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	200.00 Gbps	0.0003 s	0.0002 s	4.11	43.75%	0.498
FULL_PATH	200.00 Gbps	0.0003 s	0.0002 s	5.08	70.83%	0.507

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4x4 torus planes, so routing evaluates the two planes separately

- the wrap edges use a fixed best-twist offset, so deterministic DOR is intentionally not exposed for this topology
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 289.77 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.16 ms at p95 with link-utilization CV 0.498.
- Structural span is diameter 5 and average hops 4.07.

3D-Torus 2x4x3

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4x3 torus planes | 48 Union | 192 SSU
Exchange Node Count	24
Ssu Count	192
Union Count	48
Internal Link Count	384
Backend Link Count	144
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one 2x4x3 torus plane, and the two Union chips inside ea

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU

Field	Value
Sparse Active Ratio	0.25
Sparse Target Count	2

Routing Diversity Snapshot

SHORTEST_PATH vs DOR: 4.91x average path diversity (9.83 vs 2.00 end-to-end paths per inter-exchange SSU pair); FULL_PATH reaches 10.00 on average with peak 10 paths.

Mode	Avg Paths	Peak Paths
DOR	2	2
SHORTEST_PATH	9.826	48
FULL_PATH	10	10

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	DOR Throughput	132.64
Sparse M-to-N	DOR Throughput	266.67

Structural Metrics

Metric	Value
Diameter	6
Average Hops	4.178
Bisection Bandwidth Gbps	9,600.00
Bisection Bandwidth Gbps Per Ssu	100

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	132.64 Gbps	0.0461 s	0.0158 s	4.18	100.00%	0.493
SHORTEST_PATH	198.96 Gbps	0.0307 s	0.0158 s	4.18	100.00%	0.288
FULL_PATH	66.32 Gbps	0.0922 s	0.0158 s	5.10	100.00%	0.440

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	266.67 Gbps	0.0002 s	0.0002 s	4.16	100.00%	0.856
SHORTEST_PATH	266.67 Gbps	0.0002 s	0.0002 s	4.16	80.56%	0.530
FULL_PATH	266.67 Gbps	0.0002 s	0.0002 s	5.16	96.67%	0.521

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4x3 torus planes, so routing evaluates the two planes separately
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 198.96 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.16 ms at p95 with link-utilization CV 0.530.
- Structural span is diameter 6 and average hops 4.18.

3D-Torus 2x4x3 Best Twist

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4x3 torus planes | 48 Union | 192 SSU | axis0=[0, 2, 0] | axis1=[0, 0, 0] | axis2
Exchange Node Count	24
Ssu Count	192
Union Count	48
Internal Link Count	384
Backend Link Count	144
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one 2x4x3 torus plane, with a fixed best-twist wrap offset

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	SHORTEST_PATH Throughput	298.44
Sparse M-to-N	SHORTEST_PATH Throughput	266.67

Structural Metrics

Metric	Value
Diameter	5
Average Hops	3.927
Bisection Bandwidth Gbps	19,200.00
Bisection Bandwidth Gbps Per Ssu	200

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	298.44 Gbps	0.0205 s	0.0158 s	3.93	100.00%	0.031
FULL_PATH	83.28 Gbps	0.0734 s	0.0158 s	4.85	100.00%	0.387

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	266.67 Gbps	0.0002 s	0.0002 s	3.97	66.67%	0.572
FULL_PATH	266.67 Gbps	0.0002 s	0.0002 s	4.98	97.22%	0.539

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU

- the two Union chips inside each exchange node belong to two independent 2x4x3 torus planes, so routing evaluates the two planes separately
- the wrap edges use a fixed best-twist offset, so deterministic DOR is intentionally not exposed for this topology
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 298.44 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.16 ms at p95 with link-utilization CV 0.572.
- Structural span is diameter 5 and average hops 3.93.

3D-Torus 2x4x2

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4x2 torus planes | 32 Union | 128 SSU
Exchange Node Count	16
Ssu Count	128
Union Count	32
Internal Link Count	256
Backend Link Count	96
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one 2x4x2 torus plane, and the two Union chips inside ea

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH

Field	Value
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Routing Diversity Snapshot

SHORTEST_PATH vs DOR: 4.27x average path diversity (8.53 vs 2.00 end-to-end paths per inter-exchange SSU pair); FULL_PATH reaches 8.00 on average with peak 8 paths.

Mode	Avg Paths	Peak Paths
DOR	2	2
SHORTEST_PATH	8.533	48
FULL_PATH	8	8

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	DOR Throughput	132.29
Sparse M-to-N	DOR Throughput	200.00

Structural Metrics

Metric	Value
Diameter	6
Average Hops	4.016
Bisection Bandwidth Gbps	6,400.00
Bisection Bandwidth Gbps Per Ssu	100

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	132.29 Gbps	0.0307 s	0.0107 s	4.02	100.00%	0.559
SHORTEST_PATH	198.44 Gbps	0.0205 s	0.0107 s	4.02	100.00%	0.354
FULL_PATH	76.51 Gbps	0.0531 s	0.0107 s	4.90	100.00%	0.474

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	200.00 Gbps	0.0003 s	0.0002 s	3.94	75.00%	0.953
SHORTEST_PATH	200.00 Gbps	0.0003 s	0.0002 s	3.94	51.04%	0.609

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
FULL_PATH	200.00 Gbps	0.0003 s	0.0002 s	4.81	81.25%	0.603

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4x2 torus planes, so routing evaluates the two planes separately
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 198.44 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.24 ms at p95 with link-utilization CV 0.609.
- Structural span is diameter 6 and average hops 4.02.

3D-Torus 2x4x2 Best Twist

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4x2 torus planes | 32 Union | 128 SSU | axis0=[0, 0, 1] | axis1=[1, 0, 1] | axis2
Exchange Node Count	16
Ssu Count	128
Union Count	32
Internal Link Count	256

Field	Value
Backend Link Count	96
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one 2x4x2 torus plane, with a fixed best-twist wrap offset

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	SHORTEST_PATH Throughput	328.45
Sparse M-to-N	SHORTEST_PATH Throughput	200.00

Structural Metrics

Metric	Value
Diameter	4
Average Hops	3.512
Bisection Bandwidth Gbps	12,800.00
Bisection Bandwidth Gbps Per Ssu	200

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	328.45 Gbps	0.0124 s	0.0107 s	3.51	100.00%	0.177
FULL_PATH	136.07 Gbps	0.0299 s	0.0107 s	4.39	100.00%	0.282

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	200.00 Gbps	0.0003 s	0.0002 s	3.34	43.75%	0.943
FULL_PATH	200.00 Gbps	0.0003 s	0.0002 s	4.33	54.17%	0.496

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4x2 torus planes, so routing evaluates the two planes separately
- the wrap edges use a fixed best-twist offset, so deterministic DOR is intentionally not exposed for this topology
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 328.45 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.24 ms at p95 with link-utilization CV 0.943.
- Structural span is diameter 4 and average hops 3.51.

3D-Torus 2x4x1

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4x1 torus planes | 16 Union | 64 SSU
Exchange Node Count	8
Ssu Count	64
Union Count	16
Internal Link Count	128
Backend Link Count	32
Backend Ports Per Union	4
Backend Pattern	Each Union uses 4 backend ports on one 2x4x1 torus plane, and the two Union chips inside ea

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Routing Diversity Snapshot

SHORTEST_PATH vs DOR: 2.14x average path diversity (4.29 vs 2.00 end-to-end paths per inter-exchange SSU pair); FULL_PATH reaches 6.00 on average with peak 6 paths.

Mode	Avg Paths	Peak Paths
DOR	2	2
SHORTEST_PATH	4.286	12
FULL_PATH	6	6

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	DOR Throughput	131.25
Sparse M-to-N	DOR Throughput	177.78

Structural Metrics

Metric	Value
Diameter	5
Average Hops	3.524
Bisection Bandwidth Gbps	3,200.00
Bisection Bandwidth Gbps Per Ssu	100

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	131.25 Gbps	0.0154 s	0.0056 s	3.52	100.00%	0.577
SHORTEST_PATH	196.87 Gbps	0.0102 s	0.0056 s	3.52	100.00%	0.333
FULL_PATH	102.72 Gbps	0.0196 s	0.0056 s	4.29	100.00%	0.376

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
DOR	177.78 Gbps	0.0004 s	0.0002 s	3.72	100.00%	1.131
SHORTEST_PATH	266.67 Gbps	0.0002 s	0.0002 s	3.72	91.67%	0.721
FULL_PATH	252.63 Gbps	0.0003 s	0.0002 s	4.51	100.00%	0.521

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4x1 torus planes, so routing evaluates the two planes separately
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 196.87 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.18 ms at p95 with link-utilization CV 0.721.
- Structural span is diameter 5 and average hops 3.52.

3D-Torus 2x4x1 Best Twist

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	2 independent 2x4x1 torus planes | 16 Union | 64 SSU | axis0=[0, 2, 0] | axis1=[0, 0, 0] | axis2=
Exchange Node Count	8
Ssu Count	64

Field	Value
Union Count	16
Internal Link Count	128
Backend Link Count	32
Backend Ports Per Union	4
Backend Pattern	Each Union uses 4 backend ports on one 2x4x1 torus plane, with a fixed best-twist wrap offset

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	SHORTEST_PATH Throughput	315.00
Sparse M-to-N	SHORTEST_PATH Throughput	266.67

Structural Metrics

Metric	Value
Diameter	4
Average Hops	3.27
Bisection Bandwidth Gbps	6,400.00
Bisection Bandwidth Gbps Per Ssu	200

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	315.00 Gbps	0.0064 s	0.0056 s	3.27	100.00%	0.000
FULL_PATH	112.50 Gbps	0.0179 s	0.0056 s	4.03	100.00%	0.312

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	266.67 Gbps	0.0002 s	0.0002 s	3.41	58.33%	0.586

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
FULL_PATH	266.67 Gbps	0.0002 s	0.0002 s	4.20	79.17%	0.453

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent 2x4x1 torus planes, so routing evaluates the two planes separately
- the wrap edges use a fixed best-twist offset, so deterministic DOR is intentionally not exposed for this topology
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 315.00 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.18 ms at p95 with link-utilization CV 0.586.
- Structural span is diameter 4 and average hops 3.27.

Clos

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	18 exchange nodes
Exchange Node Count	18
Ssu Count	144
Union Count	36

Field	Value
Internal Link Count	288
Backend Link Count	144
Backend Ports Per Union	4
Backend Pattern	Each exchange node keeps two independent Union planes; each Union uplinks via 4 x 400 Gb
Clos Uplinks Per Union Plane	4
Clos Total Uplinks Per Exchange Node	8

Routing And Workload Configuration

Field	Value
Routing Mode	ECMP
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	ECMP Throughput	400.00
Sparse M-to-N	ECMP Throughput	266.67

Structural Metrics

Metric	Value
Diameter	4
Average Hops	3.902
Bisection Bandwidth Gbps	28,800.00
Bisection Bandwidth Gbps Per Ssu	400

ECMP Workload Summary

Workload	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
A2A	400.00 Gbps	0.0114 s	0.0114 s	3.90	95.10%	0.000
Sparse M-to-N	266.67 Gbps	0.0002 s	0.0002 s	3.92	33.33%	0.534

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes

- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- ECMP splits Clos traffic across equal-cost shortest paths through the upper Union stage
- each exchange contributes 4 x 400 Gbps from union0 and 4 x 400 Gbps from union1 into separate Clos spine pools
- A2A per-SSU throughput reaches 400.00 Gbps with maximum backend link utilization 95.10%.
- Sparse M-to-N communication completes in 0.16 ms at p95 with link-utilization CV 0.534.
- Structural span is diameter 4 and average hops 3.90.

Dragon-Fly

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	52 shared exchange groups | 2 DF planes | 13 local 4-Union groups/plane | 52 Union/plane
Exchange Node Count	52
Ssu Count	416
Union Count	104
Internal Link Count	832
Backend Link Count	312
Backend Ports Per Union	6
Backend Pattern	Each Union uses 3 local 400 Gbps links and 3 global 400 Gbps links inside 2 Dragon-Fly plane
Server Count	26
Server Count Per Plane	13
Df Total Server Count	26

Field	Value
Df Group Count	52
Exchange Nodes Per Server	4
Unions Per Server	4
Df External Servers Per Union	3
Df Variant	DF
Df Plane Count	2
Df Exchange Count Per Plane	52
Df Union Count Per Plane	52
Df Ssu Count Per Plane	208
Df Local Topology	fullmesh
Df Local Ports Per Union	3
Df Global Ports Per Union	3

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	SHORTEST_PATH Throughput	199.52
Sparse M-to-N	SHORTEST_PATH Throughput	200.00

Structural Metrics

Metric	Value
Diameter	5
Average Hops	4.333
Bisection Bandwidth Gbps	33,600.00
Bisection Bandwidth Gbps Per Ssu	161.5

SHORTEST_PATH Workload Summary

Workload	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
A2A	199.52 Gbps	0.0666 s	0.0332 s	4.33	100.00%	0.203
Sparse M-to-N	200.00 Gbps	0.0003 s	0.0002 s	4.39	68.75%	0.667

Key Observations

- same-exchange SSU traffic stays inside the local 8 SSU + 2 Union unit and splits across all shortest local Union choices
- each exchange group shares its 8 SSUs across two independent Dragon-Fly Union planes, with union0 on plane 0 and union1 on plane 1
- routing always follows shortest Union-to-Union paths on the backend graph, so traffic may choose either plane when both planes offer the same minimum hop count
- within each plane, local 4-Union groups and inter-group Dragon-Fly links are built independently, but the two planes only meet at the shared SSUs
- if local Union connectivity is not fully bridged, traffic between groups in the same local 4-Union block can legitimately detour through the global Dragon-Fly plane
- completion and utilization metrics are evaluated with single-direction traffic accounting
- A2A per-SSU throughput reaches 199.52 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.16 ms at p95 with link-utilization CV 0.667.
- Structural span is diameter 5 and average hops 4.33.

SparseMesh S=5;N=2

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	11 shared exchange groups | 2 SparseMesh planes | 11 Union/plane | 44 SSU/plane | s=5, N=
Exchange Node Count	11
Ssu Count	88

Field	Value
Union Count	22
Internal Link Count	176
Backend Link Count	66
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one sparsemesh plane generated from the original ring ru
Sparsemesh Variant	SparseMesh-Local
Sparsemesh Plane Count	2
Sparsemesh Exchange Count Per Plane	11
Sparsemesh Union Count Per Plane	11
Sparsemesh Ssu Count Per Plane	44
Sparsemesh Offsets	[1, 2, 3]
Sparsemesh Sparsity	5
Sparsemesh Stride Count	2
Sparsemesh Sparser	False

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	SHORTEST_PATH Throughput	362.50
Sparse M-to-N	SHORTEST_PATH Throughput	200.00

Structural Metrics

Metric	Value
Diameter	4
Average Hops	3.287
Bisection Bandwidth Gbps	9,600.00

Metric	Value
Bisection Bandwidth Gbps Per Ssu	218.2

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	362.50 Gbps	0.0077 s	0.0075 s	3.29	100.00%	0.233
FULL_PATH	145.00 Gbps	0.0192 s	0.0075 s	4.11	100.00%	0.296

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	200.00 Gbps	0.0003 s	0.0002 s	3.00	45.83%	1.132
FULL_PATH	200.00 Gbps	0.0003 s	0.0002 s	3.78	39.58%	0.526

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent sparsemesh planes, so routing evaluates the two planes separately
- within each sparsemesh plane, every Union connects to offsets ±1, ±2, ±3 on the original sparsemesh ring and never crosses through intermediate SSUs
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 362.50 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.24 ms at p95 with link-utilization CV 1.132.
- Structural span is diameter 4 and average hops 3.29.

SparseMesh S=3;N=4

Topology figure stays first in the HTML dashboard; the PDF mirrors that hierarchy with compact configuration and routing tables before the smaller observation notes.

Hardware And Topology Configuration

Field	Value
Exchange Node Unit	8 SSU + 2 Union
Ssus Per Exchange Node	8
Unions Per Exchange Node	2
Internal Ssu Union Link Gbps	200

Field	Value
Backend Link Gbps	400
Union Total Ub Ports	18
Union Ports Reserved For Ssu	8
Union Ports Reserved For Frontend Nics	4
Backend Ports Available Per Union	6
Scale	13 shared exchange groups | 2 SparseMesh planes | 13 Union/plane | 52 SSU/plane | s=3, N=
Exchange Node Count	13
Ssu Count	104
Union Count	26
Internal Link Count	208
Backend Link Count	78
Backend Ports Per Union	6
Backend Pattern	Each Union uses 6 backend ports on one sparsemesh plane generated from the original ring ru
Sparsemesh Variant	SparseMesh-Global
Sparsemesh Plane Count	2
Sparsemesh Exchange Count Per Plane	13
Sparsemesh Union Count Per Plane	13
Sparsemesh Ssu Count Per Plane	52
Sparsemesh Offsets	[1, 2, 5]
Sparsemesh Sparsity	3
Sparsemesh Stride Count	4
Sparsemesh Sparser	False

Routing And Workload Configuration

Field	Value
Routing Mode	SHORTEST_PATH
Message Size Mb	4
A2A Scope	all SSUs send to every other SSU
Sparse Active Ratio	0.25
Sparse Target Count	2

Default Route Throughput

Workload	Route	Per SSU Throughput (Gbps)
A2A	SHORTEST_PATH Throughput	371.39
Sparse M-to-N	SHORTEST_PATH Throughput	266.67

Structural Metrics

Metric	Value
Diameter	4
Average Hops	3.398
Bisection Bandwidth Gbps	12,800.00
Bisection Bandwidth Gbps Per Ssu	246.2

A2A Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	371.39 Gbps	0.0089 s	0.0088 s	3.40	100.00%	0.123
FULL_PATH	115.30 Gbps	0.0286 s	0.0088 s	4.25	100.00%	0.372

Sparse M-to-N Routing Comparison

Mode	Per SSU Throughput	Completion Time	P95 Completion	Average Hops	Max Link Utilization	Link Utilization CV
SHORTEST_PATH	266.67 Gbps	0.0002 s	0.0002 s	3.27	54.72%	0.808
FULL_PATH	266.67 Gbps	0.0002 s	0.0002 s	4.21	66.67%	0.552

Key Observations

- same-exchange SSU traffic stays inside the exchange node via Union switching
- source SSU traffic evenly splits across both local 200 Gbps uplinks into the two Union planes
- inter-exchange SSU traffic is modeled as source SSU -> source Union -> backend topology -> destination Union -> destination SSU
- destination-side traffic evenly splits from the two destination Unions down to the target SSU
- the two Union chips inside each exchange node belong to two independent sparsemesh planes, so routing evaluates the two planes separately
- within each sparsemesh plane, every Union connects to offsets ±1, ±2, ±5 on the original sparsemesh ring and never crosses through intermediate SSUs
- SHORTEST_PATH splits each Union plane across all shortest Union-to-Union paths without a fixed dimension order
- A2A per-SSU throughput reaches 371.39 Gbps with maximum backend link utilization 100.00%.
- Sparse M-to-N communication completes in 0.24 ms at p95 with link-utilization CV 0.808.
- Structural span is diameter 4 and average hops 3.40.